पेटेंट कार्यालय
शासकीय जर्नल

OFFICIAL JOURNAL
OF
THE PATENT OFFICE

निर्गमन सं. 38/2026	शुक्रवार	दिनांक: 18/09/2026
ISSUE NO. 38/2026	FRIDAY	DATE: 18/09/2026

पेटेंट कार्यालय का एक प्रकाशन
PUBLICATION OF THE PATENT OFFICE

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/07/2026
(21) Application No.202621092298 A
(43) Publication Date : 18/09/2026

(54) Title of the invention : IOT-DRIVEN SMART AGRICULTURE SYSTEM WITH MACHINE LEARNING-BASED WEATHER PREDICTION

(51) International classification	:G06N 20/00, G06Q 50/02, A01G 25/16, A01G 9/24, G01W 1/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Symbiosis International (Deemed University)
Address of Applicant :Symbiosis Knowledge Village, Gram Lavale, Tal Mulshi, Pune 412115, Maharashtra, India Pune Maharashtra India
(72)**Name of Inventor :**
1)Dr. Giridhar Urkude
2)Parinita Choudhari
3)Aditi Sawarkar
4)Shravasti Shende

(57) Abstract :
ABSTRACT IOT-DRIVEN SMART AGRICULTURE SYSTEM WITH MACHINE LEARNING-BASED WEATHER PREDICTION The present invention discloses an IoT-driven smart agriculture system (100) integrated with machine learning-based weather prediction. The system (100) comprises a sensor module (110) having a soil moisture sensor (112), a DHT11 temperature and humidity sensor (114) and a smoke sensor (116) connected to a microcontroller unit (120). The collected sensor data is wirelessly transmitted through an XBee-based wireless communication module (130) using ZigBee protocol from the agricultural field to a monitoring station. A data processing unit (140) normalizes and cleans the received data. A machine learning prediction module (150) employing an XGBoost-based prediction engine (154) trained on historical weather datasets generates weather predictions. An output and display module (160) presents real-time sensor readings and predicted weather conditions to farmers. The system enables data-driven agricultural decision-making by integrating IoT sensing, low-power wireless communication and ensemble machine learning prediction. [to be published with fig.1]

No. of Pages : 28 No. of Claims : 10